

Neural Differential Utility: Accepted Adaptive Service Control and Certified Value-Gradient Learning

Anonymous manuscript for Operations Research

Abstract. Can a provider adapt an externally priced service protocol without reducing customer utility or service? We analyze persistent tiers, physical stock, costly reconfiguration, and participation at every review. Aggregate commitments identify stock under general demand distributions. A complete tree-edge criterion characterizes when a customer-neutral static protocol can be improved despite continuation participation; primitive curvature bounds quantify the gain, and absolute switching costs produce an exact friction threshold. A remaining-payment state explains why contractual memory can matter even without physical adjustment friction. Exact continuous-action benchmarks retain the accepted values of time, demand regimes, and inherited tiers relative to a reoptimized static protocol. For implementation, compatible scalar neural critics propose policies for graph-coupled, quartic, multistage problems, and rational residual certificates bound full-horizon loss. An equally informed value-only comparison, cached-preconditioned classical solvers, and shifted graph families expose both useful approximations and adverse transfer. A certified static gate prevents negative incremental deployment value. The analysis separates structural sources of accepted adaptivity from the statistical performance of the approximation method while making both independently auditable.

Keywords: service-level agreements; inventory control; persistent adjustment; customer participation; continuation participation; value-gradient learning.

Subject classifications: Inventory/production: service contracts; Dynamic programming/optimal control: stochastic control; Analysis of algorithms: approximation guarantees.

Area of review: Stochastic Models.

1 Introduction

A service provider can revise both physical capacity and the service tier attached to it. A higher tier may earn a premium and expose the provider to greater indemnity; replacing an installed tier consumes resources. Adaptation is valuable only if its gains survive the customer’s commitments and comparison with a properly optimized static alternative. The central question of this paper is therefore operational: when can an accepted adaptive protocol create value without lowering service or customer utility, which information and endogenous state create that value, and how can an approximate implementation be audited?

We study Neural Differential Utility (NDU) as joint control of physical activity and a persistent, externally priced contractual state. The tariff and accounting rule are fixed before optimization. This is an ordinary Markov decision process with an expanded state and action space, not a way to escape conventional stochastic control. The distinctive comparison holds the customer’s utility, expected fill, and physical resource cost at the levels of an optimized continuous static contract. A common retainer, when present, changes participation levels but not the reported incremental provider rewards.

The institution is an observable, metered service protocol. A provider and customer agree on the tariff, the rule governing future revisions, and the required service before operations begin. This is a useful abstraction for auditable capacity or managed-service schedules, not a claim that all real service agreements make actions observable. We consequently also solve an extension in which the customer may leave after every review. There is no private type or hidden effort in either model; the extension relaxes continued customer commitment while leaving the tariff unchanged.

Our first contribution is an exact accepted information-value decomposition over continuous contract actions. A supporting-line argument identifies physical stock from the service and cost commitments themselves. The remaining acceptance constraint is a payment budget. Separate quadratic programs solve static, time-only, and time–regime rules; a verified Riccati recursion solves the full inherited-state problem. Every optimum is attained deterministically, including against public randomized protocols. In the canonical instance, the total gain is 0.63666527. Demand-regime information generates most of it; inherited-tier feedback adds 0.01929038 after regime information is already available. These are accepted values, not increments borrowed from an unconstrained provider calculation.

The second contribution gives primitive structure with an exit option at every review. On a scalar tree, marginal reward per payment unit must be nondecreasing from each parent to its child for the reoptimized static protocol to be globally optimal in the payment-neutral class. A violated edge supplies a two-review improvement satisfying every continuation promise. This is an explicit linear-time test on primitive marginal costs, rather than a restatement in terms of unknown optimal multipliers. Smooth adjustment gives a quantitative gain bound; absolute adjustment yields an exact two-review friction threshold. A remaining-payment state reconciles continuation participation with dynamic programming and separates contractual memory from an additional demand signal. The earlier exact customer-exit comparison and complete multi-tier inaction regions remain available in the companion.

The third contribution develops and tests a full-horizon implementation certificate. A graph-coupled quartic resource model joins genuinely multistage inherited-tier control to all continuation constraints. Scalar neural potentials yield compatible actors; value-only and derivative-weighted fits receive the same scalar

oracle observations, each with an exact error bound. Training on small graphs is followed by changes in graph topology, weights, curvature, and horizon. Cached preconditioning, repeated timings, and rational certification are included in the comparison. Strong distribution shift produces adverse learned outcomes, which remain in the results. A certified static gate prevents a failed proposal from forcing a negative incremental deployment reward. The method certifies a submitted policy; it does not assume that derivative supervision or learning is always superior.

The paper follows one sequence: institution, accepted-control structure, exact economic comparisons, and certified implementation. The electronic companion contains current proofs, inaction structure, and diagnostic details. A separate, explicitly indexed historical supplement retains the full continuous-review, diffusion, constructive-approximation, and earlier computational material, without silently removing it from the review package. These extensions have different units or evidentiary scopes and do not validate the new jump-contract result by analogy.

1.1 Related literature and the incremental contribution

Inventory decisions and service contracts have long been studied jointly. Henig et al. (1997) embed an inventory model in supply-contract design, connecting contracted quantities, delivery, and replenishment decisions. Liang and Atkins (2013), Sieke et al. (2012), and Katok et al. (2008) examine inventory agreements, penalties, coordination, and review periods. Our contribution is not the observation that contract terms affect stock. We fix the externally priced menu, retain the installed tier as a costly state, and compare accepted contingent revisions with an optimized continuous static tier under the same customer commitments.

Dynamic responses under service-level agreements are also established. Hosseinifard et al. (2022) optimize dynamic stocking under ready-rate agreements. Caggiano et al. (2007) address multiechelon service-parts agreements, and Abbasi et al. (2022) analyze service levels in vendor-managed inventory. Here even the static-contract comparator can adjust stock with the regime. The incremental controls are contractual tier revisions and their information requirements, not access to a state-aware inventory policy denied to the benchmark.

Nasser and Turcic (2019) analyze temporary contract adjustment when a retailer has a private demand forecast. Their information rents and revelation problem differ from our public-state, fixed-tariff protocol. Plambeck and Zenios (2003) study incentive-efficient control with unobserved effort. We do not claim to solve those incentive constraints: our customer-exit extension imposes continuation participation, not private-information truth telling. Combined pricing and inventory optimization is likewise established (Federgruen and Heching, 1999); demand here is exogenous rather than shifted by the chosen price.

Our monotone selections use Topkis (1978); hysteretic control has predecessors including Hipp and Holzbaur (1988). Ordered hulls, occupation measures, Lagrangian bounds, and completion of squares are standard tools. The contract-specific contributions are stock identification under aggregate commitments, the accepted information decomposition, the complete continuation-compatible edge criterion, its primitive friction and curvature implications, and an acceptance-aware full-horizon audit. First-order optimality, normal-cone representations, Bregman decompositions, and strong-convexity bounds are standard convex tools (Boyd and Vandenberghe, 2004), not independent novelty claims. Their assumptions and operational

specialization are stated explicitly.

Derivative-weighted fitting follows Sobolev training (Czarnecki et al., 2017), and polynomial graph filtering follows Defferrard et al. (2016). The retained quadratic graph critic calibrates an exactly analyzable subproblem. The new compatible softplus potential addresses a nonlinear multistage objective. Neither architecture is claimed as newly invented; the contribution is equally informed fitting, continuation-constrained deployment, and full-horizon verification with all adverse outcomes visible. Forward–backward value-process methods (Ma et al., 1994; Han et al., 2018) and Markov-chain approximation (Kushner and Dupuis, 2001) support the retained continuous-time material, which is labeled separately from the new continuous-action acceptance certificate.

2 An externally priced inventory contract

There are periods $t = 0, \dots, T - 1$. The observed state is (z, q) , where z is a demand regime in a finite ordered set $\mathcal{Z}$ and $q \in [0, 1]$ is the inherited contract. Conditional demand D_z is nonnegative with finite first moment. A decision consists of stock $S \in \mathcal{S} \subset [0, \bar{S}]$ and contract $\theta \in \Theta \subset [0, 1]$. Both feasible sets are fixed compact chains; finite grids are allowed. Unused stock perishes and unmet demand is lost. Thus no inventory or backlog is carried. The next regime has exogenous transition matrix P , and the next inherited contract equals the chosen contract.

Write the expected overage and shortage as

$$O_z(S) = \mathbb{E}(S - D_z)^+, \quad L_z(S) = \mathbb{E}(D_z - S)^+, \quad C_z(S) = cS + hO_z(S) + pL_z(S). \quad (1)$$

Here $c, h, p \geq 0$ are externally fixed physical-cost coefficients. The provider receives premium $r_z\theta$, pays incremental liability $\kappa\theta L_z(S)$, pays maintenance $m\theta^2$, and pays adjustment $\lambda(\theta - q)^2$, where $\kappa, m, \lambda > 0$. Expected one-period net reward is

$$g(z, q, S, \theta) = r_z\theta - C_z(S) - \kappa\theta L_z(S) - m\theta^2 - \lambda(\theta - q)^2. \quad (2)$$

The expectation in this expression is over demand, not over the decision. Premium rates, liability coefficients, and physical-cost weights cannot be controlled. The contract is a real cash-flow commitment. Its level affects both reward and the future state; it is not an estimator of a preference parameter.

2.1 A customer-accepted master agreement

The contract is a publicly specified menu, not a penalty that the provider can change after performance is observed. Before period zero, a risk-neutral customer and provider sign a master agreement. It specifies the premium schedule $r_z\theta$, the indemnity $\kappa\theta(D_z - S)^+$, the horizon, and a contingent operating protocol. Regime, tier, stock, demand, and fulfillment are observable and contractible. At each review the provider implements the announced protocol after observing (z, q) and before demand is realized. Public randomization is permitted when specified in the agreement. The customer commits to the contingent menu, not to a new

purchase at each review. Transfers and compliance with the protocol are enforceable. No hidden effort or private type is assumed.

Let $f_z(S) = \mathbb{E}D_z - L_z(S)$ be expected units served. For a fixed service benefit $\nu > 0$, the customer's incremental one-period utility is

$$u(z, S, \theta) = \nu f_z(S) + \kappa \theta L_z(S) - r_z \theta. \quad (3)$$

Indemnity is a transfer from the provider to this customer. Maintenance and adjustment are real resource expenditures, not payments to the customer. Write $F^\pi = \mathbb{E}^\pi \sum_t \beta^t f_{z_t}(S_t)$, $D = \mathbb{E} \sum_t \beta^t \mathbb{E}D_{z_t}$, and $U^\pi = \mathbb{E}^\pi \sum_t \beta^t u(z_t, S_t, \theta_t)$. Demand is exogenous, so D is independent of the protocol. The agreement is accepted only if

$$U^\pi \geq \underline{U}, \quad F^\pi \geq \gamma D. \quad (4)$$

These are ex ante participation and expected discounted fill commitments. They are not period-by-period chance constraints or a right to renegotiate after each regime realization. A customer with such rights would require continuation participation constraints and an additional promised-utility state.

A fixed retainer A_0 can fund provider participation. Total provider profit is $A_0 + W^\pi$ and total customer surplus is $U^\pi - A_0$; the retainer does not affect policy comparisons. In the numerical study $A_0 = 6$, $\nu = 2$, and the customer's outside alternative is the best fixed-contract arrangement, including the same retainer. Both that arrangement and the accepted adaptive arrangement satisfy a zero provider outside option. We report incremental W and U so that the historical accounting remains exactly comparable.

The institution fixes the menu and designs the contingent protocol subject to acceptance. It does not claim to solve private-information screening or to endogenize every price in the menu. The unconstrained Bellman equation below is the provider relaxation and is also the problem when the acceptance constraints are slack. Section EC.1 solves the binding-acceptance problem. A lower fill rate or customer surplus in the relaxation is not presented as an improvement under the accepted agreement.

For discount $\beta \in (0, 1]$ and zero terminal reward, the Bellman equation is

$$V_t(z, q) = \max_{S, \theta} \left\{ g(z, q, S, \theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta) \right\}, \quad V_T = 0. \quad (5)$$

Finite actions make existence immediate. For compact intervals, bounded demand moments, continuity of the stage reward, and induction give a continuous value and an attained maximum. We select the greatest optimal pair when the lattice conditions below hold.

2.2 Exact representation and comparator classes

Proposition 2.1 (Expanded actions and the fixed-contract restriction). *The model in (5) is exactly the Markov decision process with state (z, q) , action (S, θ) , reward (2), and transition $P_{zz'} \mathbf{1}_{\{q'=\theta\}}$. Its value is unchanged by flattening the product action set into one action index. A frozen-contract problem is the restriction $\theta = \bar{\theta}$; its value cannot exceed the joint value when the same initial state and external reward are used. A dummy*

label that changes neither reward nor the next state does not enlarge the attainable value.

Proof. A policy in either representation assigns the same pair to each observed history, so it induces the same joint law of states, actions, and rewards. Equivalently, the two Bellman maxima enumerate the same quantities. The frozen problem optimizes over a subset of these policies. Duplicating a maximand leaves a maximum unchanged. $\square$

Optimizing over a genuine extra decision can nevertheless produce a reduced operator that is not affine in the continuation value. For fixed physical action, write

$$\mathcal{B}(v) = \sup_{\theta} \{b_{\theta} + \beta \langle P_{\theta}, v \rangle\}. \quad (6)$$

If this envelope has two different uniquely exposed slopes on an open convex set of continuation values, it cannot equal one fixed affine reward/kernel primitive there. This is a statement about eliminating a decision while retaining one primitive, not a separation from the expanded-action problem. The proof and the precise exposed-slope condition appear in Electronic Companion (EC) Section EC.1.

Nor does support-envelope geometry exclude entropic control. For $\rho > 0$, a finite full-support reference distribution P , and any bounded vector v ,

$$\frac{1}{\rho} \log \mathbb{E}_P e^{\rho \beta v} = \sup_{Q \in \Delta} \left\{ \beta \mathbb{E}_Q v - \frac{1}{\rho} \text{KL}(Q \| P) \right\}. \quad (7)$$

The optimizer is proportional to $P \exp(\rho \beta v)$. Subtracting the objective at a particular Q from the left side gives $\rho^{-1} \text{KL}(Q \| Q^*) \geq 0$. A sufficiently rich kernel-control family therefore represents this entropic operator exactly. The inventory model uses its specific contractual family; the general framework does not claim non-equivalence to all risk-sensitive or robust-control models.

3 Structure of the provider relaxation

Throughout this section, $V = V^{\text{rel}}$ denotes the provider relaxation. Accepted values are denoted W^{acc} or V^{acc} ; a relaxation property is not transferred to an accepted optimizer without checking its constraints.

Assumption 3.1 (Premium–liability compatibility). Demand is stochastically nondecreasing in z , the regime transition is stochastically monotone, and for every $z_2 \geq z_1$ and feasible S ,

$$r_{z_2} - r_{z_1} \geq \kappa \{L_{z_2}(S) - L_{z_1}(S)\}. \quad (8)$$

All other cost coefficients and the product action set are independent of z .

For a translated demand family $D_z = D_0 + z$, shortage is 1-Lipschitz in the translation. Thus $r_{z_2} - r_{z_1} \geq \kappa(z_2 - z_1)$ is a sufficient primitive condition. It allows shortage exposure to rise; premium growth must compensate for the incremental liability attached to a larger contract.

Theorem 3.2 (Monotone stock and contract). *Under Assumption 3.1, V_t is supermodular in (z, q) for every t . The set of maximizing pairs in (5) is a compact sublattice and has a greatest element. Both components of this greatest optimizer, $S_t^*(z, q)$ and $\theta_t^*(z, q)$, are nondecreasing in z and in q .*

Proof. We verify increasing differences for the full maximand, including the continuation value. In the action pair (S, θ) , the only cross term is $-\kappa\theta L_z(S)$, which has increasing differences because L_z decreases in S . In (θ, q) , the cross difference from adjustment is $2\lambda \Delta\theta \Delta q \geq 0$.

For fixed θ , $-C_z(S) - \kappa\theta L_z(S)$ has increasing differences in (S, z) . To see this without a density assumption, its stock marginal at continuity points is

$$-c - hF_z(S) + (p + \kappa\theta)(1 - F_z(S));$$

stochastic demand ordering makes this marginal nondecreasing in z . Integrating between two stock levels proves the required finite-difference inequality, including at atoms. The stage cross difference in (θ, z) is

$$\Delta\theta [\Delta r - \kappa\Delta L(S)] \geq 0$$

by (8). All remaining cross pairs involving q or z have zero stage cross difference.

Suppose V_{t+1} is supermodular in (z', q') . For $\theta_2 \geq \theta_1$, the function $z' \mapsto V_{t+1}(z', \theta_2) - V_{t+1}(z', \theta_1)$ is nondecreasing. Stochastic monotonicity here means $\sum_{z'} P_{z_2 z'} f(z') \geq \sum_{z'} P_{z_1 z'} f(z')$ for every increasing f and $z_2 \geq z_1$, equivalently ordering every upper-tail probability. It therefore makes its conditional expectation nondecreasing in z . The continuation term has increasing differences in (z, θ) and is independent of S, q . Consequently the full maximand is supermodular on the product lattice (S, θ, z, q) .

Maximizing a supermodular function over a fixed compact action lattice preserves supermodularity in the remaining coordinates: apply the lattice inequality to maximizing pairs at two parameter states, and use their meet and join as feasible pairs at the meet and join of those states. The same inequality makes the argmax a sublattice and orders its greatest elements when the parameters increase. To make the selection step explicit, take greatest maximizers a at x and b at $y \geq x$. The joint lattice inequality gives $F(a, x) + F(b, y) \leq F(a \wedge b, x) + F(a \vee b, y)$. Optimality forces equality, so $a \vee b$ is optimal at y ; greatest selection implies $a \vee b \leq b$, hence $a \leq b$. Continuity and compactness give nonempty compact argmax sets. Closure under meet and join gives their greatest elements on the finite-dimensional compact product of chains. This is the supermodular-maximization and greatest-selection form of the lattice theorem of Topkis (1978), with its existence and order hypotheses verified here. Since $V_T = 0$ is supermodular, backward induction completes the proof. $\square$

The result concerns the state variables of the stated model. It does not assert a backlog comparative static in a model that has no carried backlog. It also does not follow for an internal penalty-only objective. With terminal value zero, $p(\theta) = \theta$, $q = 1/2$, and fixed stock, that alternative objective has optimizer

$$\arg \max_{0 \leq \theta \leq 1} \{-\theta L - \lambda(\theta - 1/2)^2\} = \text{proj}_{[0,1]} \left(\frac{1}{2} - \frac{L}{2\lambda} \right), \quad (9)$$

which decreases with exposure. External premium income in (2), and specifically (8), changes this conclusion.

3.1 Stock elimination and the inherited-contract envelope

For a fixed contract, stock solves a newsvendor problem:

$$S_z(\theta) \in \arg \min_{S \in \mathcal{S}} \{cS + hO_z(S) + (p + \kappa\theta)L_z(S)\}. \quad (10)$$

On a stock interval with a continuous strictly increasing demand distribution, an interior optimum has critical fractile

$$F_z(S_z(\theta)) = \frac{p + \kappa\theta - c}{h + p + \kappa\theta}. \quad (11)$$

Boundary solutions apply when the right side is outside the attainable distribution range. For atoms or a stock grid, (10) itself, with the greatest-minimizer convention, is the definition; no continuous quantile formula is silently substituted for grid optimization.

Set $U_z(\theta) = \max_S \{-C_z(S) - \kappa\theta L_z(S)\}$ and

$$b_t(z, \theta) = r_z\theta - (m + \lambda)\theta^2 + U_z(\theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta). \quad (12)$$

Then

$$V_t(z, q) = -\lambda q^2 + \max_{\theta \in \Theta} \{b_t(z, \theta) + 2\lambda\theta q\}. \quad (13)$$

Theorem 3.3 (Exact envelope reduction). *For any demand family and any regime transition, $q \mapsto V_t(z, q) + \lambda q^2$ is convex. If Θ contains N_θ ordered points, this function is a convex piecewise-affine envelope with ordered slopes $2\lambda\theta$. Its active line selects a greatest optimal contract, and (10) supplies the corresponding stock. After stock costs and continuation expectations are computed, the envelope can be constructed in $O(N_\theta)$ arithmetic operations from the ordered slopes and queried at N_q ordered inherited-contract states in $O(N_\theta + N_q)$ operations. No global concavity of b_t is required.*

Proof. Equation (13) is obtained by expanding $-\lambda(\theta - q)^2$. A supremum of affine functions is convex. For finite Θ , insert lines in increasing slope order. Intersections with the current last line must increase along an upper hull; otherwise the last line has an empty interval of optimality and is removed. Each line is inserted and removed at most once. At an intersection choose the larger slope, which selects the greater contract. A single forward pass through ordered queries locates all active intervals. Stock elimination is exact because the continuation and adjustment terms do not depend on stock. $\square$

The operation count is an arithmetic count, not a bit-complexity statement for arbitrarily long rational numbers. A direct factored enumeration already reduces the expensive product search: on the recorded grid, it uses 2,904 rather than 37,752 Bellman action comparisons, plus 429 one-time stock-stage computations. The hull construction further avoids enumerating every contract at every inherited state. The executed hull and factored enumeration agree with exhaustive rational optimization at all 264 nonterminal states.

3.2 A quadratic bound without global concavity

Let V now allow every contract in $[0, 1]$, retaining the same stock set. Let $V^{(n)}$ restrict all contract actions to $\Theta_n = \{0, 1/n, \dots, 1\}$ but permit any initial $q \in [0, 1]$. After the first action, the inherited state lies on the grid. Define $H_r = \sum_{k=0}^{r-1} \beta^k$, with $H_0 = 0$.

Lemma 3.4 (A nonsmooth global-maximizer lemma). *Let G be finite, convex, and continuous on $[0, 1]$, let $A \geq 0$, and set $f(x) = G(x) - Ax^2 + \ell x$. If an interior point x^* is a global maximizer, every $g \in \partial G(x^*)$ equals $2Ax^* - \ell$, and $f(x^*) - f(y) \leq A(y - x^*)^2$ for every $y \in [0, 1]$. A global maximum at an endpoint is represented exactly on every endpoint-containing grid. Consequently an n -interval uniform grid loses at most $A/(4n^2)$.*

Proof. A supporting line gives $f(x^* + h) \geq f(x^*) + (g - 2Ax^* + \ell)h - Ah^2$. If the linear coefficient were nonzero, a sufficiently small interior h of its sign would strictly improve the objective. Thus it vanishes for every subgradient. The same supporting inequality at any y gives the stated bound, and a nearest grid point has distance at most $1/(2n)$. If G has a genuine kink, its subdifferential is not a singleton and that point cannot be an interior maximum of this form. No endpoint subgradient is assumed and no global concavity of f is used. $\square$

Theorem 3.5 (Continuous-contract value enclosure). *With the primitives of (2), for $r = T - t \geq 1$,*

$$0 \leq V_t(z, q) - V_t^{(n)}(z, q) \leq \frac{(m + \lambda)H_r + \beta\lambda H_{r-1}}{4n^2}. \quad (14)$$

This is also a performance-loss bound for the exact grid-optimal policy when it is implemented in the continuous-contract problem from (z, q) . It is uniform in z, q and does not require differentiability of demand distributions, global concavity, or a numerical smoothness estimate.

Proof. First consider $f(x) = G(x) - Ax^2 + \ell x$ on $[0, 1]$, where G is finite and convex and $A > 0$. If a maximum x^* lies at an endpoint, it belongs to every grid. If it is interior, take any subgradient s of G there. The supporting-line inequality gives

$$f(x) \geq f(x^*) + (s - 2Ax^* + \ell)(x - x^*) - A(x - x^*)^2.$$

An interior maximum forces the linear coefficient to vanish: otherwise a sufficiently small displacement with its sign would improve the objective. Therefore the nearest grid point loses at most $A/(4n^2)$.

For the Bellman maximization after stock elimination, U_z is convex as a supremum of affine functions of θ . Theorem 3.3, applied to continuous actions, says that $V_{t+1}(z', \theta) + \lambda\theta^2$ is convex. Consequently its contract objective has the displayed form with $A = m + \lambda + \beta\lambda$ whenever $t < T - 1$. At the final period, $A = m + \lambda$ because $V_T = 0$. Restricting just the current maximization thus loses at most the corresponding $A/(4n^2)$.

Let $E_t = \sup_{z, q} (V_t - V_t^{(n)})$. Monotonicity and the β -Lipschitz property of the Bellman operator give

$$E_t \leq \frac{m + \lambda + \beta\lambda \mathbf{1}_{\{t < T-1\}}}{4n^2} + \beta E_{t+1}, \quad E_T = 0.$$

Summing proves (14). A grid policy uses admissible continuous actions and has exactly the grid-model reward and transition law, so its continuous-model value is $V^{(n)}$. $\square$

The theorem bounds contract discretization, not physical-stock discretization or an inventory diffusion limit. It avoids interpreting a small difference between two numerical grids as a proof. In the recorded instance, the 160-interval result encloses the continuous-contract initial value in $[-4.99278573, -4.99260152]$; the exact rational endpoints are supplied with the code.

4 Identification and a unified accepted-control problem

This section supplies a common formulation for the scalar contract model, joint service portfolios, and continuation participation. Critical-fractile analysis and convex first-order optimality are established tools; the supporting-price interpretation identifies their role here. The primitive tree criterion in Section 5, rather than the optimality conditions themselves, is the structural result used to diagnose accepted adaptation.

4.1 Demand primitives and identified physical decisions

Let D_z be nonnegative demand with finite mean, on an exogenous regime process whose occupation probabilities do not depend on the policy. On a compact stock interval $I_z = [l_z, u_z]$, $l_z < u_z$, write

$$C_z(S) = cS + h\mathbb{E}(S - D_z)^+ + p\mathbb{E}(D_z - S)^+, \quad f_z(S) = \mathbb{E} \min(S, D_z), \quad \Psi_z(S) = C_z(S) - \xi f_z(S).$$

Here $\xi > 0$ is a supporting service price, not a contractual transfer chosen by the provider. Denote the demand distribution function by H_z , avoiding confusion between physical fill and resource cost. The critical level is $\tau = (p + \xi - c)/(h + p + \xi)$, with $h + p + \xi > 0$.

Proposition 4.1 (Identified stocks, boundaries, and slack commitments). *Suppose s_z uniquely minimizes Ψ_z on I_z . Then the commitments $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ almost surely at every positive-probability review, and both commitments bind. This statement includes randomized policies.*

For an interior minimizer, the one-sided derivatives satisfy

$$\Psi'_{z,-}(S) = c - p - \xi + (h + p + \xi)H_z(S-), \quad \Psi'_{z,+}(S) = c - p - \xi + (h + p + \xi)H_z(S).$$

Thus $H_z(s_{z-}) \leq \tau \leq H_z(s_z)$. At l_z the condition is $\Psi'_{z,+}(l_z) \geq 0$, and at u_z it is $\Psi'_{z,-}(u_z) \leq 0$. A strictly straddling atom or a continuously increasing crossing gives uniqueness; when no crossing lies in the feasible interval the constrained minimizer can instead be an endpoint. The condition $0 < \tau < 1$ alone does not guarantee an interior feasible quantile.

If, separately, $\Psi_z(S) - \Psi_z(s_z) \geq \alpha|S - s_z|^2$ on every I_z , then relaxed commitments imply

$$\mathbb{E}^\pi \sum_t \beta^t |S_t - s_{z_t}|^2 \leq (\delta_C + \xi \delta_F)/\alpha, \quad F^\pi \geq F^s - \delta_F, \quad C^\pi \leq C^s + \delta_C. \quad (15)$$

A sufficient smooth-demand condition is a density at least $\rho > 0$ throughout the entire feasible stock interval, with $\alpha = (h + p + \xi)\rho/2$.

Proof. The displayed derivatives follow by differentiating expected excess and shortage from the left and right. Convexity gives the endpoint and interior conditions. Sum the nonnegative gaps $\Psi_{z_t}(S_t) - \Psi_{z_t}(s_{z_t})$ under the policy's discounted occupation probabilities. Exogeneity makes the comparator use those same probabilities, while acceptance bounds their sum above by $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Uniqueness therefore identifies stock at each reachable review. Under quadratic growth the same sum is at least the left side of (15) times α . The density bound gives $\Psi_z'' \geq 2\alpha$ almost everywhere; integration and the constrained first-order condition prove the growth inequality even at an endpoint. The smooth-growth conclusion is not attributed to the atomic canonical instance. $\square$

For d stocks with arbitrary correlated demands, a concrete vector extension is $C_z(S) = \sum_j C_{zj}(S_j)$ and $f_{zj}(S_j) = \mathbb{E} \min(S_j, D_{zj})$, with a separate commitment $F_j^\pi \geq F_j^s$ for every service and a single total physical-cost cap. Choose $\xi_j > 0$ and the unique constrained marginal quantiles at levels $(p_j + \xi_j - c_j)/(h_j + p_j + \xi_j)$. On a product stock set, $\Psi_z = C_z - \sum_j \xi_j f_{zj}$ has the unique vector minimizer. Correlation is immaterial because these costs are additive expectations. Under coordinate density bounds, (15) holds with a squared Euclidean norm, numerator $\delta_C + \sum_j \xi_j \delta_{F,j}$, and the smallest coordinate curvature. Coupled stock constraints require these coordinate minimizers to be jointly feasible; the statement is not extended to an arbitrary vector service functional by assertion.

4.2 Joint resources and explicit information restrictions

On a finite exogenous tree, let $x_n \in [0, 1]^d$ be the tier vector at decision node n , $w_n > 0$ its discounted probability, and $a_n \in \mathbb{R}^d$ its net payment rates. The fixed initial tier is q_0 . Identified physical cost C^s is constant. Set

$$\mathcal{R}(x) = \sum_n w_n \{M_n(x_n) + A_n(x_n - x_{\text{pa}(n)})\} + \mathcal{G}(x), \quad (16)$$

$$\begin{aligned} W(x) &= p^\top x - \mathcal{R}(x) - C^s, & p_n &= w_n a_n, \\ X &= \{x \in [0, 1]^{Nd} : Bx \leq b, Ex = e\}. \end{aligned} \quad (17)$$

All resource functions are differentiable and convex on a neighborhood of the box. The joint function $\mathcal{G}$ includes, for example, $\sum_n w_n \zeta_n \sum_{(i,j) \in \mathcal{E}_n} v_{nij} (x_{ni} - x_{nj})^2$. This is genuinely nonseparable within a portfolio. It is not represented by a sum of univariate functions. The equations $Ex = e$ identify actions sharing the same permitted information; equality multipliers remain unrestricted, rather than being disguised as payment prices. Each continuation row of B uses discounted subtree payments. Multiplication by a positive node probability converts a conditional participation inequality to this unconditional form.

Let $\bar{x}$ embed the reoptimized static comparator, with commitments defined from that comparator and $\bar{x} \in X$. Put $g = p - \nabla \mathcal{R}(\bar{x})$. Define

$$T_X(\bar{x}) = \overline{\{s(y - \bar{x}) : s \geq 0, y \in X\}}, \quad N_X(\bar{x}) = \{v : v^\top (y - \bar{x}) \leq 0 \forall y \in X\}.$$

Lemma 4.2 (Standard first-order alternative specialized to acceptance). *An accepted policy strictly improves $W(\bar{x})$ if and only if $g \notin N_X(\bar{x})$. Equivalently, the following explicit linear program has a positive value:*

$$\max_d g^\top d \quad \text{s.t.} \quad B_i d \leq 0 \quad (B_i \bar{x} = b_i), \quad E d = 0, \quad \|d\|_\infty \leq 1, \quad d_j \geq 0 \quad (\bar{x}_j = 0), \quad d_j \leq 0 \quad (\bar{x}_j = 1).$$

For such a direction, let $s_{\max} = \sup\{s \geq 0 : \bar{x} + ud \in X \text{ for } 0 \leq u \leq s\}$. If the directional second derivative of $\mathcal{R}$ is at most L_d on this segment, then

$$W(\bar{x} + sd) - W(\bar{x}) \geq sg^\top d - L_d s^2/2. \quad (18)$$

For $L_d > 0$, take $s = \min\{s_{\max}, g^\top d/L_d\}$.

Proof. Concavity gives $W(y) - W(\bar{x}) \leq g^\top (y - \bar{x})$. Conversely every direction in the displayed polyhedral tangent cone is feasible on a short segment, and a positive derivative gives improvement. Tangent–normal polarity gives the equivalence. Twice integrating the directional curvature bound proves (18). This is the ordinary first-order test and curvature estimate for convex optimization, not a new optimality principle (Boyd and Vandenberghe, 2004). $\square$

Lemma 4.3 (Resource loss and acceptance slack). *At an optimizer x^* , suppose $\eta \geq 0$, $\xi \in \mathbb{R}^{\text{rows}(E)}$, and $v \in N_{[0,1]^{Nd}}(x^*)$ satisfy $p - \nabla \mathcal{R}(x^*) = B^\top \eta + E^\top \xi + v$ and $\eta^\top (b - Bx^*) = 0$. For every $x \in X$,*

$$W^* - W(x) = D_{\mathcal{R}}(x, x^*) + \eta^\top (b - Bx) + v^\top (x^* - x),$$

where $D_{\mathcal{R}}(x, x^*) = \mathcal{R}(x) - \mathcal{R}(x^*) - \nabla \mathcal{R}(x^*)^\top (x - x^*)$. Every term is nonnegative.

Proof. Add and subtract the linearization of $\mathcal{R}$, substitute stationarity, and use complementary slackness and $E(x - x^*) = 0$. The signs follow from convexity, feasibility, and the outward-normal definition. For a differentiable convex objective over a nonempty polyhedron, first-order optimality and the polyhedral normal-cone representation provide these multipliers; no interior Slater point is required for that representation. This standard KKT decomposition (Boyd and Vandenberghe, 2004) separates real resource inefficiency from unused customer commitments in the present model. It is not claimed as a novel algebraic identity. $\square$

5 Primitive structure under continuation participation

Continuation participation changes which customer-neutral reallocations are permissible. We give a complete scalar tree criterion and a quantitative improvement that can be checked from local primitives, without solving the defining nonlinear control program. The graph formulation above remains available for portfolios; the scalar result is not silently promoted to a graph theorem.

5.1 An edge test rather than a high-dimensional direction search

Let o be the root of a finite decision tree. Assume $a_n > 0$, $w_n > 0$, scalar continuous tiers, and an interior constant comparator $\bar{\theta}$. Define $c_n = w_n a_n$ and the accepted, payment-neutral class

$$\sum_n c_n(x_n - \bar{\theta}) = 0, \quad \sum_{j \succeq n} c_j(x_j - \bar{\theta}) \leq 0 \quad (n \neq o), \quad 0 \leq x_n \leq 1. \quad (19)$$

Here $j \succeq n$ includes n itself. Root equality holds customer utility exactly fixed; every nonroot inequality prevents the remaining expected payment from exceeding the static outside protocol. Identified fill and physical cost stay fixed. This is a subclass of weak acceptance, so every strict improvement obtained here also improves the weakly accepted problem.

Let $\mathcal{R}$ be convex and differentiable, and define the primitive marginal reward per payment unit at the comparator by

$$k_n = \frac{w_n a_n - \partial_n \mathcal{R}(\bar{\theta} \mathbf{1})}{w_n a_n}. \quad (20)$$

For scalar maintenance and centered smooth switching, $A'_n(0) = 0$ and $\nabla \mathcal{G}(\bar{\theta} \mathbf{1}) = 0$, this is $k_n = 1 - M'_n(\bar{\theta})/a_n$ at every nonroot node. At the root, subtract also $A'_o(\bar{\theta} - q_0)/a_o$. Thus the test permits nonquadratic maintenance, arbitrary finite smooth friction, and nonstationary tariffs. Initial installation is not omitted.

Theorem 5.1 (Complete continuation-compatible edge criterion). *In (19), the static protocol is globally optimal if and only if*

$$k_n \geq k_{\text{pa}(n)} \quad \text{on every nonroot edge.} \quad (21)$$

If some edge $u \rightarrow v$ has $\Delta = k_u - k_v > 0$, set $d_u = 1/c_u$, $d_v = -1/c_v$, and all other components to zero. Then $\bar{\theta} \mathbf{1} + sd$ is feasible for

$$0 \leq s \leq s_{\max} = \min\{c_u(1 - \bar{\theta}), c_v \bar{\theta}\},$$

and strictly improves the reward for all sufficiently small positive s . If $d^\top \nabla^2 \mathcal{R}(\bar{\theta} \mathbf{1} + sd)d \leq L_d$, its gain is at least $s\Delta - L_d s^2/2$. The edge test costs $O(N)$ comparisons once the primitive marginal costs have been evaluated.

Proof. For any feasible x , put $y_n = c_n(x_n - \bar{\theta})$ and $S_n = \sum_{j \succeq n} y_j$. Then $S_o = 0$ and $S_n \leq 0$ for $n \neq o$. Since $y_n = S_n - \sum_{v: \text{pa}(v)=n} S_v$, finite summation by parts gives

$$\nabla W(\bar{\theta} \mathbf{1})^\top (x - \bar{\theta} \mathbf{1}) = \sum_{n \neq o} (k_n - k_{\text{pa}(n)}) S_n. \quad (22)$$

Condition (21) makes this nonpositive for every feasible x ; concavity proves global optimality. If an edge violates it, the displayed two-coordinate transfer has $S_v = -s$ and all other nonroot subtree sums zero. It therefore satisfies every continuation inequality, not just an ex ante budget. Its derivative is $\Delta > 0$, and the box gives $s_{\max}$. The curvature bound proves the finite gain. This also proves necessity without appealing to existence of unknown optimizer multipliers. $\square$

The direction moves expected payment from a descendant review to its parent. This timing respects a

customer's later option to leave. By contrast, shifting payment from a low-price state to a high-price sibling can violate that sibling's continuation promise, even when total payment is unchanged. The edge criterion exposes precisely the order imposed by this institution. For weak root acceptance instead of equality, the same calculation adds $k_o S_o$; the complete scalar criterion is (21) together with $k_o \geq 0$.

A primitive curvature bound is obtained by summing all affected terms:

$$L_d \leq \sum_n w_n \overline{M}_n d_n^2 + \sum_n w_n \overline{A}_n (d_n - d_{\text{pa}(n)})^2 + \overline{G}_d, \quad d_{\text{pa}(o)} = 0. \quad (23)$$

Here $\overline{M}_n, \overline{A}_n$ bound second derivatives along the entire feasible segment, and $\overline{G}_d$ bounds the joint coupling curvature. The sum includes incoming and outgoing switching edges, including unchanged children of either perturbed node. If $L_d = L_0 + \lambda L_A$, an untruncated step yields a guaranteed gain $\Delta^2/[2(L_0 + \lambda L_A)]$. Finite smooth adjustment can make the gain small without necessarily eliminating it.

5.2 An exact threshold for nonsmooth switching

The contrast with absolute adjustment is substantive rather than notational. Consider a two-review chain $o \rightarrow v$, $q_0 = \bar{\theta}$, the same root payment equality and child participation constraint, and switching cost $\lambda\{\kappa_o|x_o - q_0| + \kappa_v|x_v - x_o|\}$, where $\kappa_o, \kappa_v > 0$ include their probability weights. Let k_o, k_v now exclude that absolute switching term, and let the remaining resource cost be differentiable and convex. Set

$$K = \kappa_o/c_o + \kappa_v(1/c_o + 1/c_v), \quad \lambda_c = \max\{0, (k_o - k_v)/K\}.$$

Corollary 5.2 (Friction threshold with an exit option). *The static protocol is globally optimal exactly when $\lambda \geq \lambda_c$. For $0 \leq \lambda < \lambda_c$, a sufficiently small customer-neutral transfer from the child to its parent strictly improves reward while satisfying continuation participation.*

Proof. Every feasible displacement has $d_o = s/c_o$, $d_v = -s/c_v$ for $s \geq 0$. Its right derivative at zero is $k_o - k_v - \lambda K$. A positive derivative gives improvement. A nonpositive derivative, combined with concavity on this one-dimensional feasible interval, rules out improvement everywhere, including the equality case. The threshold therefore uses primitive rates and frictions, not an uncomputed optimizer. $\square$

5.3 What the value of contractual memory does and does not mean

Proposition 5.3 (Zero-friction bound without history-specific promises). *With exogenous regimes, compact convex tiers in $[0, 1]^d$, convex stage resources, and only ex ante linear payment commitments, assume that stage resource functions and payment coefficients depend only on time and the current regime, with no other history-dependent feasibility or physical carry-over. Let $W_{\text{full}}(\lambda)$ and $W_{t,z}(\lambda)$ allow full observed history or only time and current regime. Suppose the sole inherited-tier term is $\lambda \mathbb{E} \sum_t \beta^t \|x_t - x_{t-1}\|^2$. For any time-regime optimizer π_0 at zero friction,*

$$0 \leq W_{\text{full}}(\lambda) - W_{t,z}(\lambda) \leq \lambda \mathcal{A}(\pi_0) \leq \lambda d \sum_t \beta^t.$$

Proof. At zero friction, conditional expectation of actions given (t, z_t) preserves expected linear commitments and reduces convex stage costs. Thus both zero-friction values coincide and compactness gives π_0 . Adding nonnegative friction reduces the full value, whereas using π_0 loses at most $\lambda \mathcal{A}(\pi_0)$. Compact tier bounds give the last inequality. $\square$

The proposition deliberately excludes history-specific promises. This exclusion has content: with two equiprobable histories leading to the same terminal regime, terminal utility $x - x^2$, and respective tier caps $1/4$ and $3/4$, history-dependent control earns $(3/16 + 1/4)/2 = 7/32$, whereas a common terminal tier earns $3/16$. The gap is $1/32$ even at zero friction. Both histories and their caps are public. Different earlier installed tiers can record which promise applies. The missing information concerns a contractual obligation, not a new signal about demand. Consequently no unconditional claim that continuation-constrained memory value vanishes is made.

For the continuation institution, a remaining-payment budget supplies the appropriate dynamic state. At node n , with incoming budget b and outside-protocol cap $\bar{b}_n$, choose the tier x and child budgets b_v subject to

$$a_n^\top x + \beta \sum_{v \in \text{ch}(n)} P_{nv} b_v \leq \min\{b, \bar{b}_n\}. \quad (24)$$

The Bellman state is (n, q, b) , or (t, z, q, b) when the subtree primitives and outside caps are Markov. Its continuation term is $\beta \sum_v P_{nv} V_v(x, b_v)$. Backward induction establishes equivalence with the full tree program: feasible child budgets give a feasible subtree protocol, while a feasible protocol supplies child budgets equal to its actual expected remaining payments. At zero friction the recursion does not use q , but it still uses the promise budget and the information determining the outside caps. This is the precise continuation-aware replacement for conditioning on demand regime alone.

6 What accepted adaptivity buys

The occupation program establishes a feasible improvement, but an information decomposition must impose its commitments in every class. We now solve that comparison with continuous tiers, rather than transferring an unconstrained information increment to the accepted problem. A supporting-price argument first identifies the physical policy from the commitments themselves.

6.1 The service and resource commitments identify stock

Lemma 6.1 (Physical-policy identification). *Let s_z be a fixed-stock rule and suppose that, for some $\xi > 0$,*

$$C_z(S) - C_z(s_z) - \xi \{f_z(S) - f_z(s_z)\} \geq 0 \quad (25)$$

for every feasible z, S , with equality only at $S = s_z$. Demand regimes are exogenous. Any protocol with $F^\pi \geq F^s$ and $C^\pi \leq C^s$ must select $S_t = s_{z_t}$ almost surely at every positive-probability review. Both commitments then hold with equality. For the canonical instance, $s_z = z + 2$ and $\xi = 1$ satisfy (25), for both the original stock menu and every stock in $[0, 6]$.

Proof. The expected discounted sum of the left side of (25) is nonnegative. Feasibility bounds that sum above by $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Every nonnegative summand at a reachable state must therefore vanish. Strictness identifies stock, and exogeneity gives equality of both totals. In the instance, $C_z(z+2) = 1.03+0.3z$ and $f_z(z+2) = 1.5+z$. On the half-unit menu, the smallest positive supporting gap is exactly $9/40$. The functions are piecewise linear between the demand atoms, all of which belong to that menu. The same unique minimizer consequently holds on the entire stock interval. The positive gap $9/40$ is a menu gap, not a uniform gap on continuous stocks approaching $z+2$. $\square$

This is not unconstrained stock elimination inside a constrained problem. Both the service floor and the physical-cost cap are used to prove that no other stock decision is feasible. Under looser commitments, we retain all stock choices in the occupation program.

Set $w_{tz} = \beta^t \Pr(z_t = z)$, $a_z = r_z - \kappa L_z(s_z)$, and $A = \sum_{t,z} w_{tz} a_z$. At the canonical stocks, $a_z = 0.36+1.4z$. Let $\bar{\theta}$ be the optimized continuous static tier, and put $\bar{P} = A\bar{\theta}$. The remaining acceptance condition is a payment budget,

$$P^\pi := \mathbb{E}^\pi \sum_t \beta^t a_{z_t} \theta_t \leq \bar{P}, \quad U^\pi = vF^s - P^\pi. \quad (26)$$

The accepted reward is $W^\pi = P^\pi - C^s - \mathbb{E}^\pi \sum_t \beta^t [m\theta_t^2 + \lambda(\theta_t - q_t)^2]$. Thus matching the customer's utility fixes expected net payments. The remaining economic benefit must be a reduction in maintenance and reconfiguration resources, not an unreported transfer from the customer.

6.2 Randomization and the accepted information hierarchy

We define randomization before comparing information. An ex ante public random tape R , independent of demand regimes, may select a constant tier, a time schedule, or a map of time and current regime. These classes have rules $f(R)$, $f_t(R)$, and $f_t(z, R)$, respectively. They include deterministic rules and public mixtures of them. They do not silently grant a regime-only rule access to earlier regimes through an unrecorded memory state. Full protocols may use the entire observed history; their optimum will admit a Markov implementation.

Proposition 6.2 (Derandomization on a continuous accepted menu). *Under Lemma 6.1, continuous tier feasibility $[0, 1]$, and convex maintenance and adjustment costs, public randomization does not improve the accepted static, time-only, or time-regime value. An inherited-only randomized rule with deterministic initial tier has the same accepted value as a time-only rule. For the canonical quadratic problem, a deterministic full-state rule also attains the unrestricted accepted optimum.*

Proof. Replace each restricted rule by its expectation over R , holding the regime path fixed. The resulting action remains in $[0, 1]$ and in the same information class. Expected payments do not change. Jensen's inequality decreases each maintenance cost and each adjustment cost, including the jointly randomized difference of consecutive tiers. The fixed physical policy preserves the other commitments. Without regime inputs, an inherited-only rule generates a random time schedule from q_0 and R ; the preceding argument applies. For full histories the conditional-mean construction need not preserve the displayed Markov parametrization. The deterministic Markov rule and a matching unrestricted Lagrangian upper

bound constructed below establish attainment directly. None of these statements asserts derandomization on a coarse, nonconvex tier menu. $\square$

For a multiplier $\eta \geq 0$, maximize $W^\pi - \eta P^\pi$ and add $\eta \bar{P}$. This is a different sign convention from the customer-utility multiplier in (EC.5), with the constant customer-service term removed. Write its full-state value as

$$J_t^\eta(z, q) = -A_t q^2 + B_{tz} q + D_{tz}, \quad A_T = B_{Tz} = D_{Tz} = 0. \quad (27)$$

When the maximizers are interior, backward completion of squares gives

$$\begin{aligned} K_t &= m + \lambda + \beta A_{t+1}, & L_{tz} &= (1 - \eta) a_z + \beta \sum_{z'} P_{zz'} B_{t+1, z'}, \\ A_t &= \lambda - \lambda^2 / K_t, & B_{tz} &= \lambda L_{tz} / K_t, \\ D_{tz} &= -C_z(s_z) + \beta \sum_{z'} P_{zz'} D_{t+1, z'} + L_{tz}^2 / (4K_t), & \theta_t^\eta(z, q) &= \lambda q / K_t + L_{tz} / (2K_t). \end{aligned} \quad (28)$$

Endpoint verification is essential: the affine rule must lie in $[0, 1]$ for every $q \in [0, 1]$, not just along one simulated path. We verify both endpoints in rational arithmetic for every period and regime in the canonical certificate.

The restricted classes require their own optimization. Give each reachable time–information pair one tier variable x_i . Their rewards have the form

$$W(x) = -x^\top H x + b^\top x - c_0, \quad c^\top x \leq \bar{P}, \quad H \succ 0. \quad (29)$$

Here the maintenance terms contribute $m w_i$ to the diagonal; an adjustment edge from variable j to variable i contributes its discounted transition weight times $\lambda(x_i - x_j)^2$. The initial installation cost contributes $\lambda(x_0 - q_0)^2$. There are one, eight, and twenty-two variables in the static, time, and time–regime problems. For an interior solution,

$$x = \frac{1}{2} H^{-1} (b - \eta c), \quad \eta = \max \left\{ 0, \frac{c^\top H^{-1} b - 2\bar{P}}{c^\top H^{-1} c} \right\}.$$

The matrix is not a full-state Bellman relaxation of the regime restriction: a single variable is shared by every history with the same permitted information.

Theorem 6.3 (Exact accepted information values). *For the canonical primitives, the commitments of the optimized continuous static contract, and continuous stock and tier intervals, the accepted values in Table 1 are exact rational numbers. Each value is attained without public randomization. The accepted increment from observing the inherited tier, after time and current regime have already been admitted, is 0.019290378765... The total gain over the optimized static contract is 0.636665271611...*

Proof. The exhaustive static stock-envelope calculation yields the stated $\bar{\theta}$ and stocks. Lemma 6.1 then reduces every feasible protocol, not just the proposed ones, to (26). For each restricted class, the displayed solution of (29) has every tier strictly between zero and one. Rational arithmetic verifies its stationarity

equation, nonnegative multiplier, primal feasibility, and complementary slackness. Strict concavity proves global optimality in the deterministic class; Proposition 6.2 includes its public mixtures.

In the full problem, A_t is independent of η and B_{tz} is proportional to $1 - \eta$. Forward first moments make $P^{\theta\eta}$ affine in η . Solving $P^{\theta\eta} = \bar{P}$ gives $\eta = 0.180271243733 \dots$. Rational endpoint checks verify interior maximization in every Bellman step. The affine policy is feasible, and its Lagrangian bound is equal to its reward. Forward second moments evaluate maintenance and adjustment exactly. This bounds every randomized history-dependent protocol from above and attains the bound with a deterministic Markov policy. The files record the rational values, multipliers, and policy coefficients, rather than only the decimals in the table. $\square$

The inherited-tier increment is approximately three percent of the total accepted gain. Most of the gain is due to current-regime reallocation; attributing the entire improvement to contractual memory would be incorrect. Nevertheless, inherited information has strictly positive value under the same customer commitments, not only in a provider relaxation. The continuous optimum also exceeds the certified 0.1-menu optimum by $0.011524959639 \dots$. The earlier lottery is therefore not the source of the economic improvement.

6.3 A primitive mechanism for a robust improvement

Theorem 6.4 (A customer-neutral adaptive direction). *Consider a feasible constant tier $\bar{\theta} \in (0, 1)$ with fixed stocks s_z , linear net payments $a_z\theta$, and twice continuously differentiable maintenance M near $\bar{\theta}$, with $M'(\bar{\theta}) > 0$. Suppose that at a review $t \geq 1$ two regimes have positive discounted weights w_L, w_H and $0 < a_L < a_H$. For every finite quadratic adjustment coefficient, a sufficiently small deterministic time-regime perturbation strictly improves provider reward while preserving customer utility, fill, and physical cost exactly. The conclusion holds in a neighborhood of primitives preserving these strict inequalities.*

Proof. Change only that review's tiers by εd_z , where

$$d_H = (w_H a_H)^{-1}, \quad d_L = -(w_L a_L)^{-1}, \quad d_z = 0 \text{ otherwise.}$$

Then $\sum_z w_z a_z d_z = 0$ and $\sum_z w_z d_z = 1/a_H - 1/a_L < 0$. Payments and physical decisions are unchanged in expectation. The first-order reward gain is $\varepsilon M'(\bar{\theta})(1/a_L - 1/a_H) > 0$. All changed adjustment costs are second order because the baseline tier is constant at this and the next review; the initial installation is unaffected. If $|M''| \leq L_M$ locally, the absolute quadratic cost is at most

$$\varepsilon^2 \{L_M/2 + \lambda + \beta \lambda \mathbf{1}_{\{t < T-1\}}\} \left(\frac{1}{w_H a_H^2} + \frac{1}{w_L a_L^2} \right).$$

Choose ε small enough to keep both tiers interior and make this term smaller than the positive linear gain. Continuity preserves a strictly positive margin in a sufficiently small neighborhood. For $M(\theta) = m\theta^2$, the displayed first- and second-order terms give the exact payoff difference, not just an asymptotic expansion. $\square$

This argument does not require translated demand, a coarse menu, or a lottery. It does require positive and heterogeneous net payment rates and an interior static tier with increasing maintenance cost. It proves existence of an improvement, not that every parameter perturbation has a large gain. It also explains the role of the customer constraint: concentrating a fixed expected payment budget on higher-payment regimes permits a lower average maintained tier. Reconfiguration is worthwhile when those savings exceed its resource cost.

6.4 Allowing the customer to leave at each review

The ex ante institution can be weakened without introducing hidden action or changing the tariff. At every observed regime history h at review t , let the customer's outside option be the remaining service utility of the same static contract. The provider remains committed to the announced protocol, but the customer may leave at each such review. Physical stock is still identified by the service and cost commitments. Continuation participation therefore requires

$$\mathbb{E} \left[\sum_{s=t}^{T-1} \beta^{s-t} a_{z_s} \theta_s \middle| h \right] \leq \bar{\theta} \mathbb{E} \left[\sum_{s=t}^{T-1} \beta^{s-t} a_{z_s} \middle| h \right] \quad \text{for every reachable } h. \quad (30)$$

These constraints cannot be replaced by the root participation inequality. On the finite regime tree they are linear, while maintenance and adjustment give a strictly concave quadratic objective. The canonical tree has 3,280 decision nodes. Its policy may depend on history; a promise-state representation is an alternative, not an assumption that the original three-coordinate Markov policy remains sufficient.

We solve this convex program numerically, round its deterministic tiers to rational numbers, and contract the vector slightly toward zero to enforce every continuation inequality exactly. The resulting reward is $-5.417032209823 \dots$, an improvement of $0.453078763224 \dots$ over the static contract. A rational feasible-policy and weighted-residual certificate places the interim-participation optimum less than 6.95×10^{-7} above this reward. Thus the exit option costs approximately 0.18358651 relative to full customer commitment, but does not eliminate the gain. EC Section EC.4 gives the upper-bound proof and the complete policy audit. No compensating side payment or change in the fixed menu is used to obtain this result.

7 Compatible learning with an acceptance certificate

The accepted value function supplies both a reconfiguration rule and an audit of an approximate rule. A global sup-norm residual is not the only possible audit. On the identified quadratic model, the economically relevant loss has a direct energy representation.

Theorem 7.1 (Accepted policy-loss identity). *Suppose the feasible physical policy is identified as in Lemma 6.1, and the interior Bellman recursion (28) has a nonnegative multiplier η^* whose optimal protocol satisfies $P^* = \bar{P}$. For any accepted protocol π , including a learned or randomized one,*

$$W^{\text{acc},*} - W^\pi = \mathbb{E}^\pi \sum_t \beta^t K_t [\theta_t - \theta_t^{\eta^*}(z_t, q_t)]^2 + \eta^* (\bar{P} - P^\pi). \quad (31)$$

The identity concerns continuous tier actions and inherited states, not a scan of a finite chain. Certified upper bounds on the energy and the acceptance slack give a policy-loss certificate on that continuous-action model.

Proof. Completion of squares in each Bellman maximization gives its exact action gap as $K_t[\theta - \theta_t^{\eta^*}(z, q)]^2$. Sum along the law of π ; the discounted continuation terms telescope and the terminal value is zero. The left side before adding the commitment constant is $J_0^{\eta^*,*} - (W^\pi - \eta^*P^\pi)$. Since $W^{\text{acc},*} = J_0^{\eta^*,*} + \eta^*\bar{P}$, rearrangement gives (31). No smoothness assumption on the learned policy or uniform residual bound enters the argument. $\square$

The identity does not make solving every control problem free: its coefficients must be obtained and verified, and its expectation must be bounded for the deployed policy. In the scalar affine case, first and second moments suffice. In a more general model a valid dual value or residual inequality is still needed. The next experiment uses such an inequality and does not require a tensor grid over its inherited contractual state.

7.1 A coupled portfolio at a service review

Consider a customer buying d metered services under one portfolio agreement. Each service has the same physical primitives as above, with its observed regime fixed at this review. Aggregate fill and physical-cost commitments identify stock by summing (25) over services. The net payment vector a is fixed in the menu. Let $q \in [0, 1]^d$ be the installed tier vector. In addition to maintenance $m\|\theta\|^2$, the provider incurs $\zeta\theta^\top L\theta$ to maintain different configurations across linked services, where L is the graph Laplacian of the publicly specified service-dependency graph and $\zeta \geq 0$. This is a nonseparable configuration cost, not independent copies labeled as a high-dimensional problem.

At the current review, or at a terminal Bellman update, the accepted operating problem is

$$\max_{\theta \in [0,1]^d} \{a^\top \theta - C - m\|\theta\|^2 - \zeta\theta^\top L\theta - \lambda\|\theta - q\|^2\} \quad \text{subject to } a^\top \theta \leq \bar{P}. \quad (32)$$

Here acceptance is by the common portfolio buyer, not a claim of separate participation by every service's end user. For $q = \frac{1}{2}\mathbf{1}$, the uniform static tier is optimized continuously; $L\mathbf{1} = 0$ gives

$$\bar{\theta} = \frac{d^{-1}\mathbf{1}^\top a + \lambda}{2(m + \lambda)}, \quad \bar{P} = \bar{\theta}^\top \mathbf{1}^\top a.$$

The experiment isolates a coupled, many-service accepted update. It is not presented as a 1,024-dimensional replication of the entire eight-review information experiment or as a generic nonquadratic inventory solution.

Put $K = (m + \lambda)I + \zeta L$ and $b_\eta = (1 - \eta)a + 2\lambda q$. The unconstrained quadratic Lagrangian critic is

$$J^\eta(q) = -C - \lambda\|q\|^2 + \frac{1}{4}b_\eta^\top K^{-1}b_\eta. \quad (33)$$

It is the actual operating value whenever its maximizer is interior. The training queries below satisfy that condition by the discrete maximum principle for K : its inverse is nonnegative and $K^{-1}\mathbf{1} = (m + \lambda)^{-1}\mathbf{1}$. In

the audit, an unconstrained Lagrangian remains an upper bound even if a different candidate reaches the box boundary.

For a symmetric polynomial filter $p_c(K)$, define the learned scalar critic by replacing K^{-1} in (33). Its derivative is compatible by construction, and its associated actor is

$$\widehat{\theta} = q + \frac{\nabla_q \widehat{J}^\eta(q)}{2\lambda} = \frac{1}{2} p_c(K) b_\eta. \quad (34)$$

The computation is a graph-filter network with fixed message-passing features and a trained scalar quadratic readout. Polynomial graph filters are established constructions (Defferrard et al., 2016); the new use here is accepted tier control with an independent deployment certificate, not a claim to have invented graph convolution or polynomial inversion.

Theorem 7.2 (A computable accepted deployment bound). *Let $K \succeq \mu I$ with $\mu > 0$. For any rationally specified feasible tier vector $\widehat{\theta}$, any $\eta \geq 0$, and $r = (1 - \eta)a + 2\lambda q - 2K\widehat{\theta}$,*

$$0 \leq W^{\text{acc},*} - W(\widehat{\theta}) \leq \frac{\|r\|^2}{4\mu} + \eta(\overline{P} - a^\top \widehat{\theta}). \quad (35)$$

Thus rational input coefficients, deployed tiers, and multiplier yield an exact arithmetic upper bound without solving the optimal program or sampling the inherited-state domain.

Proof. Weak Lagrangian duality bounds the accepted optimum by the unconstrained maximum of $b_\eta^\top \theta - \theta^\top K \theta$ plus its constants. Expanding about $\widehat{\theta}$ leaves $r^\top d - d^\top K d \leq \|r\|^2/(4\mu)$. Adding the nonnegative commitment slack gives (35). Maximizing over all real d only enlarges the upper bound, so the argument does not require an interior optimal box solution. $\square$

This bound is an ex post guarantee for the specified operational instance and deployed vector. It is not a distribution-free prediction of neural performance on an unseen graph. Each unseen graph is audited separately. Unlike the earlier finite-chain residual budget, the arithmetic of this audit is exact, and its scale can be compared directly with the operational improvement.

7.2 Equal information, transfer, and classical baselines

We fix $m = 1.5$, $\lambda = 0.6$, and degree-12 Chebyshev features on the spectral interval $[2.1, 26.1]$. Graphs consist of a cycle and a random matching, so their maximum degree is at most three; $\zeta \in \{1, 2, 3, 4\}$ gives the stated spectral enclosure. The network has thirteen trained readout coefficients, independent of the number of services. Its feature layers require sparse matrix–vector products, not a learned dense inverse.

Each of ten training seeds supplies 96 independently generated graphs of sizes 32 or 64. For each graph, we make exactly three scalar teacher queries at $q - hu$, q , and $q + hu$, with $h = 0.025$ and a random sign vector u . We use the operational value (33) at $\eta = 0.2$, after removing its known $-C - \lambda\|q\|^2$ terms. Both methods receive all 288 scalar observations. Value-only fitting minimizes their squared errors. Value-gradient fitting adds the directional derivative discrepancy obtained by differencing those same endpoint values. There

are no extra teacher values or derivative-oracle calls. Values are normalized by graph size and directional differences by its square root; both objectives use the same least-squares readout solve.

Because both the teacher and the learned scalar critic are quadratic in q , the centered directional difference is exact in real arithmetic. The gradient objective is therefore also an explicit finite-difference regularization of the same three value observations. We do not count that algebraically identical formulation as another independent baseline. We test $h = 0.01, 0.05, 0.1$ separately. Sampling seeds quantify variation in the training data; there is no nonconvex optimizer seed to conceal behind this convex readout fit. The scalar construction, the previously compiled cubic-ReLU envelopes, and the historical trained tanh network are distinct implementations.

We deploy without retraining on twelve held-out graphs: four each at 64, 256, and 1,024 services. For each graph, two filtered right-hand sides determine the multiplier that meets the payment budget. Tiers and the multiplier are rounded to denominator 10^{10} ; a one-sided correction enforces the customer budget exactly. All coordinates, every commitment, and (35) are then checked using integer and rational arithmetic. A failed certificate would remain a failed deployment, not a discarded seed.

Classical comparisons include the analytic degree-12 Chebyshev inverse, dense Cholesky solution, sparse direct solution, and conjugate gradients with relative residual tolerance 10^{-4} . All solve the same accepted problem, including multiplier calibration. The graph representation is common input to all methods. Training time is reported separately; execution times do not include graph construction or the independent rational audit. Consequently a faster actor is not described as a faster end-to-end proof of optimality.

On the 1,024-service cases, the learned value-gradient actor has a median deployment time of approximately 1.21 milliseconds, versus 7.62 milliseconds for sparse direct solution and 18.01 milliseconds for dense Cholesky in the recorded environment. Conjugate gradients take approximately 1.52 milliseconds and are more accurate. The largest exact loss bound for the learned actor is 8.05×10^{-7} per service, whereas its smallest guaranteed gain over the static alternative exceeds 0.01606 per service. These are decision-useful enclosures, not bounds larger than the economic comparison. The same thirteen coefficients transfer to new graph sizes and graph structures.

The derivative-weighted fit lowers the mean realized loss modestly relative to the equal-information value-only fit, but improves six of ten paired training seeds at each test size, not every seed. The seed-level uncertainty is reported in the companion. The operational conclusion is not universal superiority of gradient weighting: it is that a compact learned compatible critic can deliver accepted decisions with a small, independently proved loss at a nontrivial coupled scale. The conjugate-gradient and analytic-filter results remain visible so that the contribution is evaluated against numerical linear algebra, not only against a deliberately weak neural baseline.

Re-audit of the quadratic timing comparison. The preceding retained timings factorized the classical operator inside each call and excluded the independent audit. They therefore do not establish an amortized repeated-solve advantage. R10 separately tests cached sparse and dense factorizations, Jacobi-preconditioned iteration, the analytic filter, and the same learned readout on the identical recorded portfolio instances, with 15 repetitions and end-to-end audit costs (Table 6 in the companion). Setup is separated and all exact accuracy

bounds remain visible. This correction changes the interpretation of the original timing comparison, not its historical observations or certificates.

8 Certified learning for nonlinear multistage service control

The quadratic portfolio study tests an exactly analyzable deployment primitive. A multistage contract requires more: the action changes inherited tiers, all future reviews enter the objective, and participation must hold on every continuation subtree. We now study this full finite-tree problem with nonquadratic resources. The experiment isolates approximation from verification; it does not assume that learning dominates a classical optimizer.

8.1 A common nonlinear objective and a cumulative certificate

Let $D = \text{diag}(w)$ repeat each node's discounted probability for every service. On a binary scenario tree, let $x_n \in [0, 1]^d$, $q_0 = 1/2$, and set

$$F(x) = \sum_n w_n \{m \|x_n\|^2 + \gamma \sum_j x_{nj}^4 + \lambda \|x_n - x_{\text{pa}(n)}\|^2\} + \zeta \sum_n w_n \sum_{(i,j) \in E} \omega_{ij} (x_{ni} - x_{nj})^2, \quad J(p) = \max_{0 \leq x \leq 1, Ax \leq b} \{p^\top Dx - F(x)\}. \quad (36)$$

Here $m > 0$, $\gamma > 0$, and graph weights are nonnegative. Every row of A gives total discounted tariff payment in one subtree; its cap equals the corresponding payment of the static outside protocol. Therefore every observed history has a participation constraint. Spatial graph edges and temporal parent–child edges appear simultaneously. All scenarios and parameters are specified before operations, while a tree action depends only on its realized history; no future realization is revealed to the controller.

The net tariff is a , and $p = a + r$ collects a linear provider resource saving r . Adding a fixed resource offset of ten per service–review makes the simulated resource costs nonnegative without changing any decision or reward difference. This extension does not represent endogenous pricing. In each central test instance, weighted-centered rational shocks set $p_i = m + \gamma/2 + e_i$ with $\sum_i w_i e_i = 0$. Hence the derivative of the constant-tier objective at one half is exactly zero. Strict concavity makes that tier the unique reoptimized static optimum, including initial installation; it is feasible for every continuation constraint. The comparison does not select a weak baseline to favor the learner.

Theorem 8.1 (Full-horizon nonlinear residual certificate). *For any feasible x in (36) and any $\eta \geq 0$, define*

$$r = Dp - \nabla F(x) - A^\top \eta, \quad \varepsilon(x, \eta) = \frac{1}{4m} \sum_i \frac{r_i^2}{w_i} + \eta^\top (b - Ax).$$

Then $0 \leq J(p) - [p^\top Dx - F(x)] \leq \varepsilon(x, \eta)$. This is a cumulative discounted policy-value bound for the whole contingent tree, not a single-review error. With rational primitives and rational x, η , feasibility and the bound are exactly checkable, including quartic, graph, and all parent–child terms.

Proof. Since $\nabla^2 F \succeq 2mD$, for any feasible y the reward difference is at most

$$r^\top(y - x) - m\|y - x\|_D^2 + \eta^\top A(y - x) \leq r^\top(y - x) - m\|y - x\|_D^2 + \eta^\top(b - Ax).$$

Completing the square bounds the first two terms by $\|r\|_{D^{-1}}^2/(4m)$. Taking the supremum over y proves the assertion. No floating-point solver status or learned generalization premise enters this inequality. $\square$

This is the standard strong-convexity residual argument (Boyd and Vandenberghe, 2004) specialized to the continuation-constrained nonlinear tree. Its operational content is an independently replayable full-horizon upper bound. Learning supplies a candidate, not the validity of that bound.

The same upper bound supports a no-harm deployment gate. Let V_x and V_s be the exact rewards of the proposed and optimized static feasible protocols. Deploy whichever has larger reward; its upper gap is

$$V_x + \varepsilon(x, \eta) - \max\{V_x, V_s\} \geq 0. \quad (37)$$

Thus failed transfer cannot force adoption of a worse protocol. For a prescribed loss tolerance, a candidate whose bound is too large must be refined by a classical solver until a valid certificate meets that tolerance; the experiments do not count such refinement as free. The present timing table measures candidate construction and audit, not a hypothetical unmeasured equal-accuracy hybrid.

For a receding-horizon implementation, the budget state in (24) must be propagated. If a feasible action and child-budget allocation incur a verified Bellman loss at most ϵ_t relative to the same continuation value, the standard Bellman telescoping argument gives cumulative loss at most $\mathbb{E} \sum_t \beta^t \epsilon_t$. This observation requires valid continuation bounds; separate one-step neural residuals alone do not establish it. The experiment below instead certifies complete contingent plans directly by Theorem 8.1.

8.2 Nonlinear compatible critics and equally available scalar observations

Strong convexity yields a unique $x^*(p)$ and the envelope identity $\nabla_p J(p) = Dx^*(p)$. Moreover, ∇J is Lipschitz from the D norm to its dual with constant $1/(2m)$. We fit a scalar neural potential and obtain its actor by differentiation, rather than fitting unrelated value and policy outputs. The potential uses 63 softplus features of the forcing residual and its first two local graph-tree smoothing transforms. The transforms are D -self-adjoint; the trained parameters are the readout coefficients. This finite-feature neural model is nonlinear and compatible, but is not claimed to be a new universal architecture.

Each of 24 training instances contributes the same three scalar observations at $p - hu, p, p + hu$, with $h = 1/40$ and a sign direction u . The value-only and value-gradient objectives use exactly those 72 values. Endpoint perturbations hold the tariff, outside protocol, and feasible set fixed: they are local value-function oracle queries, not new reoptimized static comparisons. Both fits use the same fixed ridge, features, and scaling. Unlike the earlier quadratic diagnostic, centered differences here are not exact derivatives. If the two endpoint value errors are bounded by ϵ_- and ϵ_+ , the directional target for the value residual, normalized

by $M = \sum_i w_i$, satisfies

$$\left| \widehat{\partial}_u(J - J_s)/M - \partial_u(J - J_s)/M \right| \leq \frac{h\|u\|_D^2}{4mM} + \frac{\epsilon_- + \epsilon_+}{2hM}. \quad (38)$$

Indeed, integrate the directional derivative between the endpoints and use the Lipschitz constant; J_s is linear in p . All 720 scalar training records have rationally verified error bars. The largest unnormalized oracle gap is 6.143×10^{-8} in the recorded run, and the largest normalized directional error bound is approximately 0.004167. These are target-error guarantees, not a statistical guarantee for the fitted network.

A deployment first clips the actor to the box, rounds down to rational tiers, and applies a common nonnegative scaling that satisfies every subtree cap. Multipliers are proposed by nonnegative least squares and rounded; an independent verifier recomputes all derivatives and participation inequalities with fractions. The static gate then compares exact values. No failed or negative-reward proposal is discarded from the reported sample.

8.3 Design and results under graph, horizon, and curvature shifts

Ten training seeds use three-review trees and four-service cycle graphs, or 28 tier coordinates. Testing uses two separately generated instances in each of four families: the training-sized cycle; an eight-service random-edge graph; an eight-service geometric graph on a four-review tree; and a weighted two-block graph with 32 services on a five-review tree. The largest contingent plan has 992 coordinates. The last two families also change quartic curvature and coupling strength. Thus topology, degree, weights, horizon, and operator spectrum can change together. This is a deliberately difficult synthetic transfer study, not evidence of calibration to a real service operation.

Classical comparisons solve exactly the same nonquadratic tree problem. Newton–direct factorizes its changing Hessian. Newton–PCG caches one baseline Hessian factorization per problem and uses it as a preconditioner for all subsequent changing Hessians and repeated solves. Setup is separated from online cost. Both methods retain the common participation dual, rational repair, and full audit. Fifteen repetitions follow one warmup; raw timing observations, setup costs, and training times are retained. Timing is not pooled with training-seed uncertainty.

Table 3 reports every outcome. All 176 deployment candidates satisfy all continuation constraints after repair, with 2,640 exact subtree checks and 52,624 exact tier-bound checks. Classical maximum normalized upper gaps are below 5.7×10^{-10} . Learned proposals create positive mean incremental reward in the cycle, random-edge, and geometric families, but both learned methods lose reward on average under the severe weighted-block shift. The static gate preserves nonnegative incremental reward without concealing those raw failures. The neural upper bounds on this shifted family are correspondingly loose; the earlier one-millionth quadratic bound is not transferred to this experiment.

Across the eight problems, the ten paired training-seed mean differences favor derivative weighting by approximately 0.000577 per discounted service–review, with a Student 95% interval of $[-0.003923, 0.005076]$. The interval includes zero. These observations do not establish a reproducible superiority claim for derivative supervision. They do establish that the accepted nonlinear multistage implementation, its adverse transfer

outcomes, and its complete value certificates can be examined independently.

Table 4 includes candidate generation, repair, multiplier proposal where needed, and rational certification. Cached preconditioning materially reduces the measured cost of the largest classical case relative to repeated changing-Hessian factorization. Neural candidate times are smaller in this run, but their accuracy is different and some proposals fail the economic improvement test. Neither this table nor the quadratic portfolio table implies a repeated-solve advantage at matched certified loss.

9 Conclusions and operational implications

The relevant benchmark is an optimized accepted static protocol, not an unoptimized tier or a provider relaxation. Under continuation participation, primitive marginal reward ratios reveal which intertemporal reallocations remain acceptable. Smooth and nonsmooth frictions have different implications for strict gain, and a promise budget can remain relevant after physical switching friction vanishes. These structural conclusions identify where adaptivity can matter before a learning method is selected.

Implementation requires a separate standard of evidence. The nonlinear tree certificates apply to whole contingent plans and are independent of how those plans were proposed. Learned critics can provide useful candidates, but their strong-shift failures and inconclusive seed intervals preclude a universal superiority claim. A static gate and a declared certified-loss tolerance turn those limitations into explicit deployment decisions rather than hidden experimental exclusions. A calibrated service-system application, endogenous tariffs, and private information are important further research directions; none is claimed to have been established by the present synthetic, observable-state experiments.

Data, software, and computation accessibility

The versioned review package supplies all source code, rational policies, oracle error records, raw timing observations, and independent replay scripts. The electronic companion specifies the entry points and distinguishes recorded runs from environment-dependent reruns. All historical scientific files remain available in the separate historical supplement and prior revision directories. No confidential or personal data are used.

References

- Abbasi B, Bruzda J, Gavirneni S (2022) Optimal operational service levels in vendor managed inventory contracts—an exact approach. *Operations Research Letters* 50(5):610–617. doi:10.1016/j.orl.2022.08.007.
- Boyd S, Vandenberghe L (2004) *Convex Optimization* (Cambridge University Press, Cambridge).
- Caggiano KE, Jackson PL, Muckstadt JA, Rappold JA (2007) Optimizing service parts inventory in a multiechelon, multi-item supply chain with time-based customer service-level agreements. *Operations Research* 55(2):303–318. doi:10.1287/opre.1060.0345.
- Czarnecki WM, Osindero S, Jaderberg M, Swirszcz G, Pascanu R (2017) Sobolev training for neural networks. *Advances in Neural Information Processing Systems* 30. arXiv:1706.04859.
- Defferrard M, Bresson X, Vandergheynst P (2016) Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in Neural Information Processing Systems* 29.
- Federgruen A, Heching A (1999) Combined pricing and inventory control under uncertainty. *Operations Research* 47(3):454–475.
- Han J, Jentzen A, E W (2018) Solving high-dimensional partial differential equations using deep learning. *Proceedings of the National Academy of Sciences* 115(34):8505–8510.
- Henig M, Gerchak Y, Ernst R, Pyke DF (1997) An inventory model embedded in designing a supply contract. *Management Science* 43(2):184–189. doi:10.1287/mnsc.43.2.184.
- Hipp SK, Holzbaur UD (1988) Decision processes with monotone hysteretic policies. *Operations Research* 36(4):585–588. doi:10.1287/opre.36.4.585.
- Hosseiniifard Z, Shao L, Talluri S (2022) Service-level agreement with dynamic inventory policy: The effect of the performance review period and the incentive structure. *Decision Sciences* 53:802–826. doi:10.1111/deci.12506.
- Katok E, Thomas D, Davis A (2008) Inventory service-level agreements as coordination mechanisms: The effect of review periods. *Manufacturing & Service Operations Management* 10(4):609–624. doi:10.1287/msom.1070.0188.
- Kushner HJ, Dupuis P (2001) *Numerical Methods for Stochastic Control Problems in Continuous Time*, 2nd ed. (Springer, New York).
- Liang L, Atkins D (2013) Designing service level agreements for inventory management. *Production and Operations Management* 22(5):1103–1117. doi:10.1111/poms.12033.
- Ma J, Protter P, Yong J (1994) Solving forward-backward stochastic differential equations explicitly—a four step scheme. *Probability Theory and Related Fields* 98:339–359.

- Nasser S, Turcic D (2019) Temporary contract adjustment to a retailer with a private demand forecast. *Management Science* 65(1):209–229. doi:10.1287/mnsc.2017.2942.
- Plambeck EL, Zenios SA (2003) Incentive efficient control of a make-to-stock production system. *Operations Research* 51(3):371–386. doi:10.1287/opre.51.3.371.14954.
- Sieke MA, Seifert RW, Thonemann UW (2012) Designing service level contracts for supply chain coordination. *Production and Operations Management* 21(4):698–714. doi:10.1111/j.1937-5956.2011.01301.x.
- Topkis DM (1978) Minimizing a submodular function on a lattice. *Operations Research* 26(2):305–321.

Table 1: Accepted continuous-tier information values

Information	Provider reward	Increment	Payment price
Static	-5.870110973	—	0.000000000
Time	-5.869294096	0.000816877	0.001568294
Time–regime	-5.252736080	0.616558016	0.174849751
Full state	-5.233445701	0.019290379	0.180271244

All four classes have the same filled demand 18.021386714, customer utility 28.657506145, and physical cost 9.587377732. All policies are deterministic. Decimals round exact rational values; stock is optimized over the full interval through Lemma 6.1.

Table 2: Accepted coupled portfolios: 1,024 persistent tier coordinates

Method	Mean loss/service	Max. bound/service	Time (ms)
Learned value–gradient	1.52e-08	8.04e-07	1.214
Learned value only	1.68e-08	9.01e-07	1.222
Analytic Chebyshev	2.69e-08	3.67e-07	1.220
Conjugate gradient	6.11e-10	3.46e-09	1.524
Sparse direct	0.00e+00	7.80e-12	7.619
Dense Cholesky	0.00e+00	7.80e-12	18.010

All deployed policies satisfy customer acceptance exactly. Learned rows aggregate ten predeclared training seeds on four fixed test graphs; other rows use those same four graphs. Loss is against the sparse-direct value. Bounds certify the rounded actions in rational arithmetic. Time is the median recorded two-right-hand-side solve plus acceptance calibration, excluding offline training and rational audit. Small direct-method loss is rounded to zero, not asserted identically zero against the unknown real-arithmetic optimum.

Table 3: Nonlinear multistage control: mean normalized gain and exact audit. Gains divide total discounted reward by discounted service-reviews. Each learned row averages ten training seeds and two problem seeds; a classical row has two problems. The maximum certificate bound is not an estimated realized loss. “Static gate” selects the higher exact reward of the proposal and optimized static rule; no test is removed.

Family	Method	Raw gain	Static gate	Max. bound	Gate returns static
Cycle	Newton-direct	0.011549	0.011549	5.63e-10	0/2
Cycle	Newton-PCG	0.011549	0.011549	5.63e-10	0/2
Cycle	Value only	0.008528	0.008528	0.0124	0/20
Cycle	Value-gradient	0.009686	0.009686	0.00784	0/20
Random edge	Newton-direct	0.014342	0.014342	1.6e-10	0/2
Random edge	Newton-PCG	0.014342	0.014342	1.6e-10	0/2
Random edge	Value only	0.012158	0.012158	0.0221	0/20
Random edge	Value-gradient	0.013160	0.013160	0.00649	0/20
Geometric	Newton-direct	0.009155	0.009155	4.56e-10	0/2
Geometric	Newton-PCG	0.009155	0.009155	5e-10	0/2
Geometric	Value only	0.002601	0.002634	0.0397	2/20
Geometric	Value-gradient	0.003773	0.003829	0.0287	1/20
Weighted block	Newton-direct	0.001953	0.001953	5.51e-10	0/2
Weighted block	Newton-PCG	0.001953	0.001953	5.51e-10	0/2
Weighted block	Value only	-0.097068	0.000000	4.8	20/20
Weighted block	Value-gradient	-0.098095	0.000000	3.69	20/20

Table 4: Measured end-to-end candidate construction, feasibility repair, and rational audit, in milliseconds. Entries average the two problem-instance medians and their 10th–90th percentiles, each from 15 repetitions after one warmup. Learned timing uses the prespecified training seed zero, not the fastest seed. Factorization setup and training are separate. Methods do not have matched accuracy; this table is not a speedup-at-equal-quality claim.

Family	Method	Median	[10th, 90th]
Cycle	Newton–direct	14.89	[14.82, 15.00]
Cycle	Newton–PCG	22.21	[22.06, 22.44]
Cycle	Value only	6.39	[6.35, 6.55]
Cycle	Value–gradient	6.36	[6.33, 6.40]
Random edge	Newton–direct	15.09	[14.97, 15.27]
Random edge	Newton–PCG	19.21	[19.10, 19.38]
Random edge	Value only	9.87	[9.82, 9.97]
Random edge	Value–gradient	9.83	[9.79, 9.90]
Geometric	Newton–direct	39.55	[39.36, 39.75]
Geometric	Newton–PCG	48.37	[48.10, 48.69]
Geometric	Value only	17.88	[17.81, 18.05]
Geometric	Value–gradient	18.00	[17.92, 18.13]
Weighted block	Newton–direct	506.82	[505.12, 508.47]
Weighted block	Newton–PCG	269.91	[268.29, 271.84]
Weighted block	Value only	214.51	[213.73, 216.34]
Weighted block	Value–gradient	215.54	[214.32, 217.20]